CS-1101
PROGRAMMING FUNDAMENTALS

Lab 09

By

Engr. Muhammad Awais

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Lab Contents

- Control Structures
- Repetition Control Structures
  - While Loop

Lab Objectives

- The objective of this lab is to learn how to use loops in C++ to perform repetitive tasks efficiently.

Need of Loop

- In programming, sometimes there is a need to perform some operation **more than once or (say) n number of times.**
- For example, suppose we want to print **"Hello World" 5 times.** Manually, we must write cout for the C++ statement 5 times as shown

```
#include <iostream>
using namespace std;
int main() {
    // Print hello world 5 times
    cout << "Hello World\n";
    cout << "Hello World\n";
    cout << "Hello World\n";
    cout << "Hello World\n";
    cout << "Hello World\n";
    return 0;
}
```

Need of Loop

- Let's say you have to write it **20 times** (it would surely take more time to write 20 statements).
- Now imagine you have to write it 100 times, it would be really hectic to re-write the same statement again and again. **In this case, loops come into use allowing users to repeatedly execute a block of statements.**

```
#include <iostream>
using namespace std;
int main() {
    int i=0;
    while(i<5){
        cout << "Hello World" << endl;
        i++;
    }
    return 0;
}
```

What are Loops ?

- A loop is an instruction that is used to **repeatedly execute a code until a certain condition is reached**.
- Loops can be classified on the basis of the sequency in which they check the condition relative to the execution of the associated block of code.
  - **Entry Controlled loops**: In this type of loop, the test condition is tested before entering the loop body.
  - **Exit Controlled Loops:** In this type of loop, the test condition is tested or evaluated at the end of the loop body. Therefore, the loop body will execute at least once, irrespective of whether the test condition is true or false.

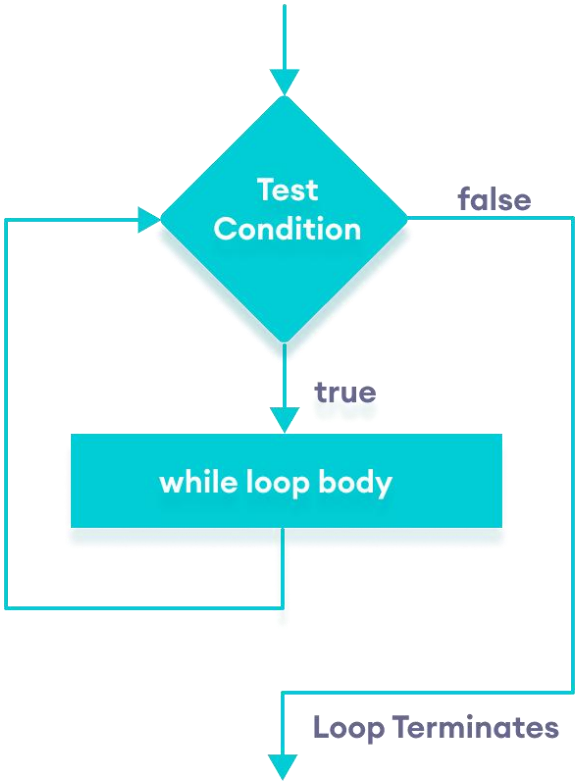
Types of Loop?

- while
- do-while
- for

While Loop

- It is an **entry-controlled** loop whose execution is terminated based on the test conditions.

```
while (condition) {
// body of the loop
update expression
}
```



Loop that will run forever

```
#include <iostream>
using namespace std;
int main() {
    while(1) {
        cout << "This loop will run forever.\n";
    }
    return 0;
}
```

```
This loop will run forever.
This loop will run forever.
This loop will run forever.
```

Tasks of while loop

- Write a program to print all numbers from 1 to N, where N is entered by the user.
- Write a program to display the multiplication table of a number entered by the user.
- Write a C++ program that repeatedly asks the user to input a positive number. If the user enters a negative number, zero, the program should display an appropriate error message and ask again. Use a while loop to ensure the program keeps asking until the user enters a valid positive number. Once a positive number is entered, display the number and terminate the program.
- Write a C++ program to calculate the sum of the first five natural numbers (1, 2, 3, 4, 5) using a while loop

Tasks of while loop

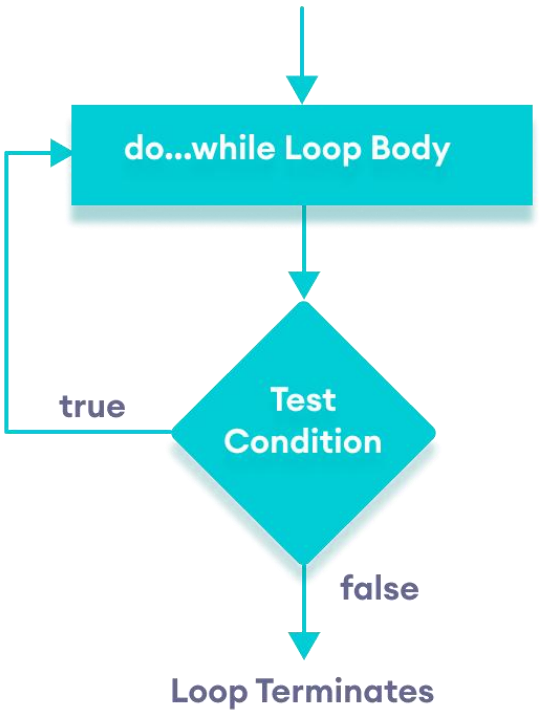
- Implement a menu-driven program for a simple banking system. The program should allow the user to perform the following operations:
  - **Deposit Money:** Add a specified amount to the account balance.
  - **Withdraw Money:** Deduct a specified amount from the account balance, ensuring that the balance does not go negative.
  - **Check Balance:** Display the current account balance.
  - **Exit:** Terminate the program.

  The system should initialize with a default balance (e.g., $1000) and **should continue to display the menu until the user chooses to exit**. Ensure the program handles invalid inputs gracefully, such as insufficient balance during withdrawal or invalid menu choices.

do-while loop

- It is an **exit-controlled** loop.
- The **do-while** loop is a variant

of the while loop with one important difference: the body of do-while loop is executed once before the condition is checked.




```
do
{
// body of loop;
} while (condition);
```

Tasks of do-while loop

- Write a program that prompts the user to enter a password. If the entered password is incorrect, they should be asked to try again until the password matches the correct one.
- Write a program for restaurant which provides user the facility to order items from the menu provided .User should be able to select from menu and submenu listed under each main menu item. Price for each item is fixed/given by District Food Authority. Your Program should be able to calculate the total bill , the user have to pay and the ability to order until he wants. Develop your program according to the requirements table given in next slide. User should be able to tell the amount he wants to order for item .For example (if he wants 2 KG of chicken karahi etc…)

Tasks of do-while loop

Main Menu Items	Sub Menus
Breakfast	• Anda Paratha (RS.40) • Nan Channy (RS)60) • Siri Paye (RS.150) • Tea (RS.15)
Lunch	• Chicken Karahi (RS.1050/KG) • Mutton Karahi (RS.1800/KG) • Egg Fried Rice(RS.450/palte) • BBQ (RS.1050/dozen)
Dinner	• Chicken Karahi (RS.1050/KG) • Mutton Karahi (RS.1800/KG) • Egg Fried Rice(RS.450/palte) • BBQ (RS.1050/dozen) • Saji (RS.800/KG)

References

- https://www.geeksforgeeks.org/cpp-loops/